

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTEPC302 - ELECTRONICS DEVICES & CIRCUITS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
-

Q1. Objective type questions. (Compulsory Question)

12

1. What does JFET stand for?
 - A. Junction Forward Emitter Transistor
 - B. Junction Field Effect Transistor
 - C. Junction Feedback Emission Transistor
 - D. Junction Field Emission Transistor
2. The JFET has how many terminals?
 - A. Two
 - B. Three
 - C. Four
 - D. Five
3. Which of the following is a voltage-controlled device?
 - A. BJT
 - B. JFET
 - C. SCR
 - D. UJT
4. A differential amplifier amplifies:
 - A. Only one input
 - B. Common mode signals
 - C. The difference between two input signals
 - D. The sum of two input signals
5. In an inverting amplifier configuration, the input is applied to:
 - A. Non-inverting terminal
 - B. Inverting terminal
 - C. Ground
 - D. Output terminal
6. Which feedback topology provides low input and low output impedance?
 - A. Voltage-series
 - B. Voltage-shunt
 - C. Current-series
 - D. Current-shunt

- 7 A typical RC phase-shift oscillator has a phase shift of:
 A. 90°
 B. 180°
 C. 360°
 D. 270°
- 8 Crystal oscillators are mainly used for:
 A. Power amplification
 B. Fixed and accurate frequency generation
 C. Audio frequency generation
 D. Phase-shift applications
- 9 The basic function of a voltage regulator is to:
 A. Amplify AC signals
 B. Filter noise
 C. Maintain constant output voltage
 D. Step up voltage
- 10 A transducer is a device that:
 A. Stores energy
 B. Converts one form of energy to another
 C. Amplifies signals
 D. Filters noise
- 11 A thermistor is a:
 A. Voltage controlled resistor
 B. Temperature dependent resistor
 C. Magnetic sensor
 D. Optical sensor
- 12 LDR stands for:
 A. Light Dependent Relay
 B. Light Dependent Resistance
 C. Low Dynamic Range
 D. Light Detection Resistor
- Q2. Solve the following.
- A) Describe the **construction and working** of an n-channel JFET with the help of a neat diagram. 6
- B) Describe the construction of an n-channel E-MOSFET. 6
- Q3. Solve the following.
- A) With a neat diagram, explain the dual input balanced-output differential amplifier. 6
- B) Explain following term 6
 a) scaling amplifier
 b) voltage follower
- Q4. Solve Any Two of the following.
- A) Explain R C phase shift oscillator in detail. 6
- B) Write short note on Wien Bridge oscillator. 6
- C) Write short note on colpitts oscillator 6
- Q5. Solve Any Two of the following.

- | | | |
|----|---|---|
| A) | With a neat circuit diagram, explain the working of a discrete transistor series voltage regulator. | 6 |
| B) | Explain IC Voltage Regulators in details | 6 |
| C) | Write short note on series voltage regulator | 6 |

Q6. Solve Any Two of the following.

- | | | |
|----|--|---|
| A) | Define a sensor and a transducer . What is the difference between them? | 6 |
| B) | What is an LDR (Light Dependent Resistor)? Explain in detail | 6 |
| C) | Write short note on | 6 |
| | a) Thermistor | |
| | b) Thermocouple | |

*** End ***